4th grade

Hi Scientist!

This is **NOT** a test.

We just want to learn about **YOU** so we can make **Science on the Court** super fun!

Do your best and show us how you think!

✓ Take your time ✓ Show your work ✓ Ask questions if you need help

Problems 1 & 2: Multi-Step Problems and Number Theory

🧩 #1: Multi-Step Word Problem

A basketball league scenario: 12 teams in the league, 9 players on each team, 15 players are absent today.

12 teams × 9 players = ? Total players

Subtract 15 absent

Players present = ?

How many players are PRESENT at practice?

❑ **A)** 93 ❑ **B)** 103 ❑ **C)** 108 ❑ **D)** 123

Show All Steps:

Step 1: _____

Step 2: _____

Final Answer: _____

💡 **Strategy**: Multiply first, then subtract!

🔢 #2: Prime and Composite Numbers

These jersey numbers are available:

Prime number = has ONLY two factors (1 and itself).
Composite = has MORE than two factors

Which jersey number is a PRIME number?

❑ **A) 9** (factors: 1, 3, 9) ❑ **B) 11** (factors: 1, 11)
❑ **C) 15** (factors: 1, 3, 5, 15) ❑ **D)** All are prime

Explain What Makes a Number Prime:

_____

_____

List factors of #11: _____

💡 **Tip**: Check all factors!

Problems 3 & 4: Fraction Operations

÷ #3: Comparing Fractions

Free throw competition: Team A made 3/4 of their shots, Team B made 5/8 of their shots

Team A: $\frac{3}{4}$

Team B: $\frac{5}{8}$

Which team made a GREATER fraction of their free throws?

- ☐ **A)** Team A (3/4)
- ☐ **B)** Team B (5/8)
- ☐ **C)** They're equal
- ☐ **D)** Cannot determine

Tip: Use common denominators!

Show Your Comparison:

Common denominator: _____

$\frac{3}{4} = \frac{___}{8}$

$\frac{5}{8} = \frac{___}{8}$

Which is greater? _____

× #4: Multiplying Fractions

Water consumption: Each team drinks 1/2 gallon of water, 3 teams are practicing today

$3 \times \frac{1}{2} = ?$

How many gallons do the 3 teams drink IN TOTAL?

- ☐ **A)** 1/2 gallon
- ☐ **B)** 1 gallon
- ☐ **C)** 1 1/2 gallons
- ☐ **D)** 2 gallons

Tip: 3 halves = ?

Show Your Calculation:

$3 \times \frac{1}{2} =$

_____ = _____

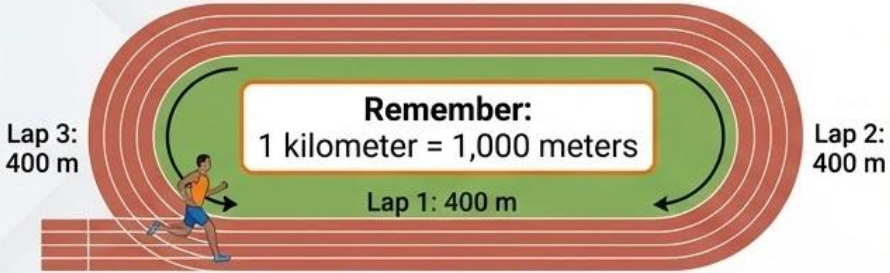
Explain your strategy:

Problems 5 & 6: Measurement and Geometry

#5: Converting Measurements

Story

Track practice: The track is 400 meters long.
A runner completes 3 full laps.



How many KILOMETERS did the runner complete?

- ☐ **A)** 0.12 kilometers
- ☐ **B)** 1.2 kilometers
- ☐ **C)** 12 kilometers
- ☐ **D)** 120 kilometers

Show Your Conversion:

Total meters: _____

÷ 1,000 = _____ km

Explain: _____

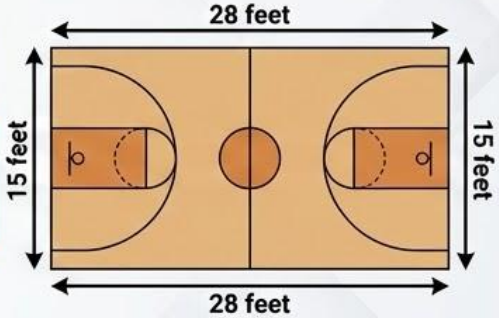
Tip: Divide by 1,000!

#6: Perimeter of the Court

Story

A rectangular basketball court:
LENGTH: 28 feet,
WIDTH: 15 feet.

Perimeter = 2(L + W)
or
Perimeter = L + W + L + W



What is the PERIMETER of the court?

- ☐ **A)** 43 feet
- ☐ **B)** 58 feet
- ☐ **C)** 86 feet
- ☐ **D)** 420 feet

Show Your Calculation:

Method: _____

P = _____

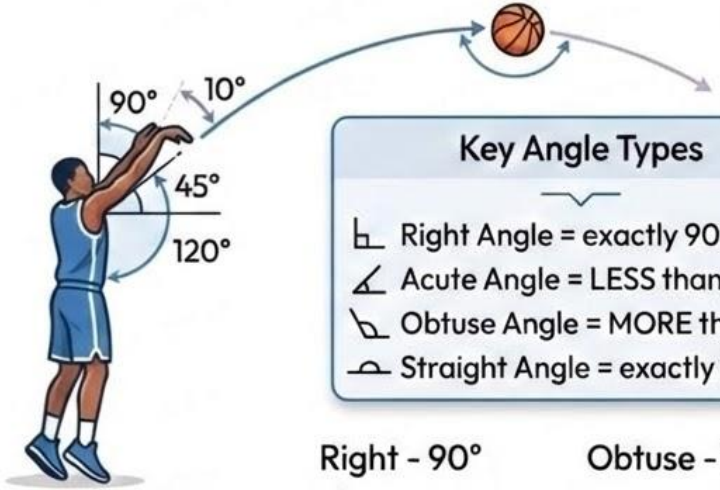
P = _____ feet

Tip: Add all 4 sides!

Problem 7: Angle Measurement

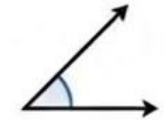
Problem 7: Understanding Angles

When a player shoots a free throw, their arm makes different angles.



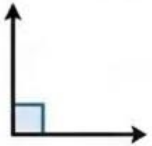
Key Angle Types

- Right Angle = exactly 90°
- Acute Angle = LESS than 90°
- Obtuse Angle = MORE than 90°
- Straight Angle = exactly 180°



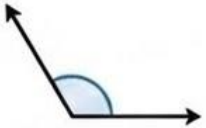
Acute - 45°

Right - 90°



Right - 90°

Obtuse - 120°



Obtuse - 120°

? If an angle measures 45°, what TYPE of angle is it?

- ☐ A) Right angle (exactly 90°)
- ☐ B) Acute angle (less than 90°)
- ☐ C) Obtuse angle (more than 90°)
- ☐ D) Straight angle (exactly 180°)

Explain Your Reasoning:

How do you know 45° is this type of angle?

Compare: Is 45° bigger or smaller than 90°? _____

0° ----45°----- 90° ---------- 180°

Tip: Compare the angle to 90° (right angle)!